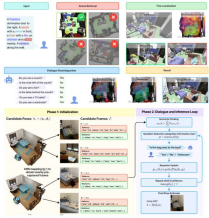


LangLoc: "Tell Me What You See" (Jul, 2026)



Performs coarse -> fine -> portion of the SLAM pipeline without images in a digital twin environment
This is done through the generation of a matching scene through semantic interview with the user (An interactive Bayesian refinement module that resolves ambiguous descriptions through targeted yes/no questions.)

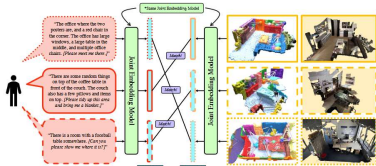
In this no images are transmitted saving both privacy and bandwidth. First of its kind for performing fine localization.

This is instead done by generatively creating scene and ranking their match against an existing digital twin.

This could be a good source of models for future implementations.

"Where am I?" Scene Retrieval with Language (2024)

Precursor to the above paper.

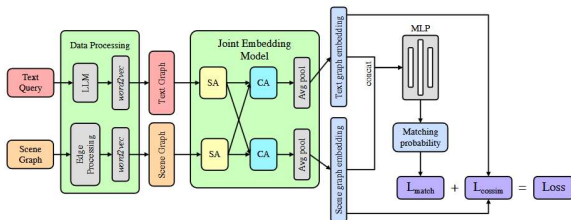


Pair embeddings are one for a text description of the scene using Word2Vec (500 D) and another for the visual portion. These are compared and used to train a model on a "room" by room basis.

These embeddings go into a self/cross attention layer to pull out similarity. This is done for each room to localize.

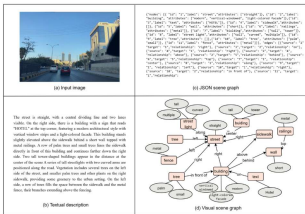
Text2SceneGraph

7



Novel to this paper ^^^^ Text2SceneGraphMatcher (Text2SGM)

Text2Graph VPR: A Text-to-Graph Expert System for Explainable Place Recognition in Changing Environments (2025)



To convert textual descriptions into structured semantic representations, we generate scene graphs where nodes denote static objects in the environment and edges represent spatial relationships between them. By converting language into structured symbolic graphs, this stage enables rule-like reasoning and supports explicit knowledge representation.

The main issue here is that GPT-4 is the language engine. CAN OTHER MLLMs HAVE SUFFICIENT COMPLEXITY TO GENERATE SUFFICIENT DESCRIPTIONS?

Approach is similar to what the previous paper seems like it would do. Scene graph -> embedded vector -> similarity comparison across vectors.

One modification. Individual scene graphs are merged into a location graph. Matching is done against the location graph and not the individual scene graphs.

SceneGraphLoc: Cross-Modal Coarse Visual Localization on 3D Scene Graphs

Slightly finer localization than Text2Graph VPR by using patched.

SG2Loc: Sequential Visual Localization on 3D Scene Graphs (June 2026)

Semantic graphs are used as one of the final steps in a multimodal localization algorithm. SceneGraphLoc is part of the pipeline

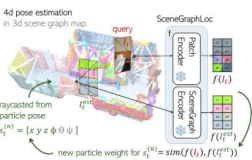


Figure 1. Observation model for a particle $s_t^{(n)} = (x, y, z, \phi)$, matching object descriptors predicted by (Miao et al., 2024) from the query image I_t to projected object labels from the semantically segmented scene mesh. The particle weight reflects how well the observed and projected objects agree.

This uses a semantic graph except is paired against a rough 3D mesh. (Potential expansion model that is not private)

Potential Gap
Adding text OCR information to semantic scene graphs to improve performance in ISE
Adapting any of this to be used on a phone.

Strong Alternative
Routing of VPR between model complexities to

Key Issue
The generation of the semantic graphs requires extraction of visual relationships into a visual form. This is done well by VLLMs but is computationally expensive and VLLMs cannot run on mobile.

Dedicated lower resource pipelines such as RT-DETRv2-M -> EGTR may be a solution. They do have a small vocabulary though.

RT-DETRv2-M is 100G FLOPS. An iPhone 13 can perform 1.3T FLOPS. At 1 frame or a few key frames per second this is reasonable.

<https://github.com/lyuwenyu/RT-DETR>

Takeaways

Your 3D map does not need to have any visual information. It can be purely semantic relationships. The use of a transformer can turn an image of a scene into a semantic relationship and then compare that across the semantic graphs from the map.

CHECK IF THIS HAS BEEN APPLIED TO ISE?

Text is lighting condition immune. Seasons will not affect matching a chair to a chair like descriptors.

"Robustness to Environmental Variations: Text abstracts away low-level pixel details and emphasizes high-level semantic content. As a result, descriptions remain relatively stable across varying lighting conditions, seasonal changes, dynamic objects, and occlusions, enhancing place recognition robustness in challenging real-world scenarios." [Paper](#)

Graph based matching system are incredibly resource efficient.

"Efficiency in Storage and Retrieval: Graph-based embeddings derived from compact textual inputs require significantly less storage and bandwidth than dense visual features and enabling faster retrieval. Moreover, similarity computation in the graph-semantic space is efficient and less sensitive to visual redundancy." [Paper](#)

Map -> Text Connection (Potentially useful in visual impairment navigation)

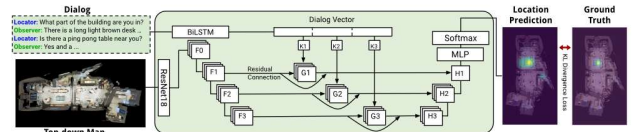
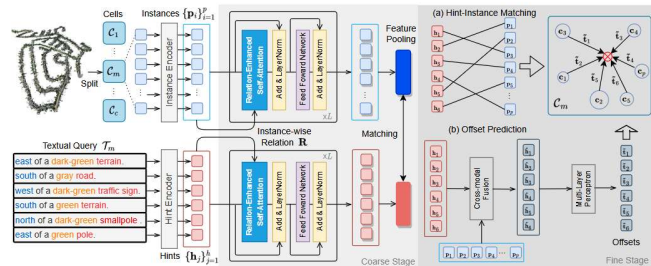
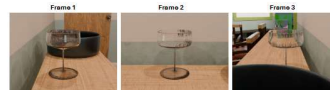


Figure 5: The 3-layer LingUNet-Skip architecture used to model the Localization from Embodied Dialog task.

Where Are You? Localization from Embodied Dialog (2021)



Text to Point Cloud Localization with Relation-Enhanced Transformer (Jan 2023)



Q: What is the spatial relationship between the bowl and the wineglass in Frame 2?
A: In Frame 2, the bowl is not visible. However, using the other frames to understand the layout of the scene, we can determine their spatial relationship.
- In Frame 5, we can see both objects. The large black bowl is located behind and slightly to the right of the stemmed glass.
- In Frame 3 is a direct, frontal view of the stemmed glass.
- Therefore, in the context of the scene as viewed in Frame 2, the bowl is positioned directly behind the stemmed glass, which is why it is completely obscured from view.

